# 1. Operations Planning

## 1.1. Operations decisions

1. Link with marketing
   - Operations manager needs to know market demand forecasts to be able to match supply and demand. This is known as operations planning.
   - If sales forecasts are accurate:
     - Easily match supply to demand
     - Keep inventory levels to a minimum efficient level
     - Reduce wastage
     - Employee appropriate number of factors of production
     - Produce the right product mix
2. Availability of resources
   - Production of goods and services requires – land, labour, capital, raw materials
   - Lack of these will influence operations decisions:
     - Location – locate in areas with abundant supply of materials
     - Nature of production method – if labour productivity is high, business may use labour intensive production method
     - Automation – if technology is cheaper, business may decide to switch to automated production method.
3. Technology
   - Technological developments have changed the production process
   - They help the process become more efficient and cost effective

## 1.2. The Transformational Process

Definition: An activity or group of activities that transform one or more input, adds value to them, and produce outputs for customers.

It takes the following inputs:

- Enterprise
- Land
- Capital
- Labour

Adds value to them via production, whether capital or labour intensive.

Transform them into the following output:

- Finished goods
- Services
- Components for other firms

## 1.3. Contribution of Operations

**Contribution of operation to improve added value:**

- Efficiency: Minimize production costs to stay competitive by improving efficiency.
- Quality: Ensure goods or services meet their intended purpose.
- Flexibility & Innovation: Adapt to new processes and products.

Amount of added value depends on inputs and different factors.

**The following factor affecting Added Value are not operational management issues:**

- Design of Products: Make it cost-effective to produce while having high-quality visual appeal to justify a higher price.
- Operational Efficiency: To cut waste and boost productivity to lower per unit costs and increase value.
- Branding: Create strong branding to make consumers willing to pay more than the production cost, like with luxury clothing, stationary etc.

## 1.4. Efficiency Production leading to Effective Sustainability

One of the main goal of an operations manager is resource management. they plan to optimize resource use by being efficient in product and minimize negative impacts on future generations through compliance to sustainability.

One major misunderstanding occurs through knowing the difference of productivity and production. The difference is:

- Production is the measure of the total output in a given period.
- Productivity is the measure of how inputs are converted into outputs per time period.

There is an calculation of calculating productivity and that is by using the labour/Capital productivity equation.

Labor productivity (number of units per worker) = Total Output in a given time / Total Workers Employed

Capital productivity (number of units per worker) = Total Output in a given time period / Total Capital Employed

There are also ways to raise productivity and they are:

- Boost Skills: Training improves productivity but is costly and risks losing staff.
- Enhance Motivation: Financial and non-financial incentives can motivate employees and cut costs. Although Non-financial incentives are usually preferred as it does not raise labour cost.
- Upgrade Equipment: New and improved technology increases output. However, it requires large investment and retraining for employees.



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- Improve Management: Effective management can raise productivity level by handling resources and workers effectively.

**However, high productivity does not guarantee success as:**

- Wage demand: Increased effort for higher productivity might cause workers to seek higher pay, which could cancel out the productivity gains.
- Worker Resistance: Workers might be reluctant to follow necessary steps to raise productivity. Improving productivity by 20% could cause job losses and potential industrial disputes if sales don't grow.
- Management Role: If the management quality is poor, success is unlikely. Therefore, high quality management that involves workers and values their input can improve productivity and acceptance.
- Efficiency or Effectiveness?: Despite raising productivity, it might not lead to success as productivity is measured via efficiency and not rather effectiveness.

Effectiveness should be focused on as well, it means meeting the needs of the customers profitably. They, combined together, gives the best outcome. Efficiency is focused on reducing the average cost of production while effectiveness is ensuring that the product that is being produced meets the needs of their targeted audience profitably.

### Sustainability of operation

It is done via maintaining business operations long term by focusing on environmental protection and preserving quality of life for future generations.

**Ways to operate business sustainably:**

- Using recycled materials
- Producing goods that can be recycled
- Waste management in production
- Buying sustainable resources from suppliers
- Reducing energy usage and carbon emission
- Lowering the usage of non-biodegradable materials such as plastic

**Why business wants sustainability in their operation:**

- To comply with strict laws on environmental issues
- Pressure group activity exposing environment-damaging businesses
- Businesses should follow through on corporate responsibility promises through their senior management.
- Sustainable practices can improve public relations, enhancing positive publicity.
- Consumers are more likely to buy eco-friendly products, increasing sales.

| Benefits of Sustainability | Drawbacks of Sustainability |
|---|---|
| Lower Energy Costs: Using less energy saves money. | Going green may require costly investments like solar panels |
| Fall in production usage and sales of plastic and non-biodegradable materials draws eco-conscious buyers | Eco-friendly materials might be pricier and less effective than plastics |
| Using recycled materials reduces need for new raw materials. Thus, potentially decreasing costs | Recycled materials often need extra cleaning or processing, heightening production completion time |
| Recyclable products can cut waste disposal cost | Creating recyclable products is likely to be costly and time-consuming |
| Lowering waste from operations decreases overall production costs | It might require Investing in worker training and advanced equipment |
| Purchase of resources from sustainable suppliers supports sustainability and reduces bad publicity risk | Sustainable supplies to make product might be more expensive, raising overall costs |
| | It might not be suitable for business target audience income |

## 1.5. Need for flexibility and innovation

- Flexibility is the business's ability to vary production with changes in demand
- Ways to increase flexibility –
  - Increase capacity
  - Hold higher stocks
  - Have a flexible workforce
  - Flexible flow production equipment

### Process innovation

- It involves the use of new, advanced technology to improve production
- Done through using CAM, CAD, robots, faster machines, computer tracking inventory system, etc
- Gives a competitive edge
- Better quality
- Higher reputation and brand loyalty
- Expensive



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## 1.6. Labour Intensive vs Capital Intensive

| Benefits of Labour Intensive Production | Benefits of Capital Intensive Production |
|---|---|
| Low machine expense | Economics of Scale |
| Interesting and varied work thus higher employee motivation | Consistent quality |
| One-off design/Job production processed product can be made that meets customer requirements precisely. | Low average cost of production |
| Lower start up cost as buying numerous machines to begin operation increase cost of a business by a great margin | Higher ability to supply to Mass market |

| Drawbacks of Labour Intensive Production | Drawback of Capital Intensive Production |
|---|---|
| Lower output level in comparison | High fixed cost |
| Skilled, High-paid workers might be required for overall operation | Cost of financing the machinaries or finding sufficient finance might be time-consuming |
| Quality of the product made depends on experience, motivation and skill of each worker | High maintenance cost and higher cost for repairs as skilled workers are necessary |
| | Due to constant improvement in technology. It leads to latest equipment depreciating fast and becoming obsolete. |

The approach of the production method that should be chosen depends on:

- Nature of the product
- Brand Image
- Relative cost of Labour and Capital
- Size of Business
- Accessibility to finance

## 1.7. Economies of scale

- Cost benefits arising with increased scale of operations

### Types of economies of scale

1. Purchasing economies
   - Bulk buying economies
   - Suppliers may offer discounts on bulk purchases
   - They will want to keep large customers happy so may provide good quality goods and on time delivery
2. Technical economies
   - High output will lower unit costs
   - Fixed costs are spread across the output, lowering its output
3. Financial economies
   - Banks and other financial institutions will be willing to provide loans to larger businesses
   - They may be willing to charge lower interest rates to them
4. Marketing economies
   - Costs of advertising and promotion maybe spread over a larger output, lowering unit costs
5. Managerial economies
   - Employing specialists and managers will be easier for large firms as their salary will be spread over a larger output

## 1.8. Diseconomies of scale

- Factors which lead to a rise in average costs of production arising with increased scale of operations beyond a certain size.

### Types of diseconomies of scale

1. Communication problems
   - In a large firm, the feedback provided will be poor
   - The chain of command may be long leading to distortion of messages
   - This may cause poor decision making
2. Alienation of the workforce
   - The bigger the organisation, the more difficult it becomes to involve every worker.
   - They may feel demotivated due to lower job satisfaction
3. Poor coordination

### How to avoid diseconomies of scale

1. Management by objectives: This will help avoid coordination problems
2. Decentralisation: This gives divisions a considerable degree of autonomy and independence.
3. Reduce diversification: Businesses that concentrate on 'core' activities may help to reduce coordination problems and some communication problems.



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## 1.9. Production methods

1. Job production
   - Producing a single, one-off item, specifically designed for the customer
   - It is labour intensive
   - Specific consumer needs are met, higher reputation and loyalty
   - Specialised products are produced, ability to charge higher prices
   - Cannot enjoy economies of scale
   - Expensive as requires skilled labour and training
2. Batch production
   - Producing products in separate groups where the entire groups goes through the production process together
   - Enables division of labour and specialisation
   - Economies of scale
   - Easy to alter batches according to demand and preferences
   - High storage costs – work in progress inventory
   - Workers may get bored and demotivated
3. Flow production
   - Producing products in a continuous process through the use of technology
   - Higher output
   - Economies of scale
   - Consistent and standardised quality
   - Low labour costs
   - High initial costs
   - Lower job security
4. Mass customisation
   - It involves the use of computer aided production to meet specific customer needs at mass production costs
   - Allows businesses to focus on differentiated marketing
   - Increases added value
   - Low unit costs
   - Customer needs are met

## 1.10. Production methods – making the choice

1. Size of market – if the market is small, flow production can not be used, batch or job production is more appropriate
2. Amount of capital available – employing flow production is expensive and requires a high initial capital investment. Small firms may not be able to afford this and therefore use job or batch production
3. Availability of other resources – using flow production requires a high supply of unskilled workers and huge land area. Job production requires highly skilled workers. The chosen production method may even depend on whether the company is able to allocate these resources.
4. Market demand exists for products adapted to specific customer requirements – if the company wants low costs but has a differentiated target market, mass customisation is the best option.

### Problems of changing production methods

- Job to batch: high equipment costs, need for extra working capital and fall in employee morale
- Job/batch to flow: high capital cost, costs of employee training, need for accurate demand forecasts

### Conclusion: Production method

Traditional differences between production methods are becoming less obvious. Technology allows large businesses to meet the diverse range of customer needs, which could threaten small firms that focus on niche markets with specialized products. However, as consumers become wealthier, there will still be demand for unique and specialized items, so small firms can continue to thrive in the niche market regardless of dominance of large businesses in terms of market share.



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